Medium Access Control (MAC) Protocols

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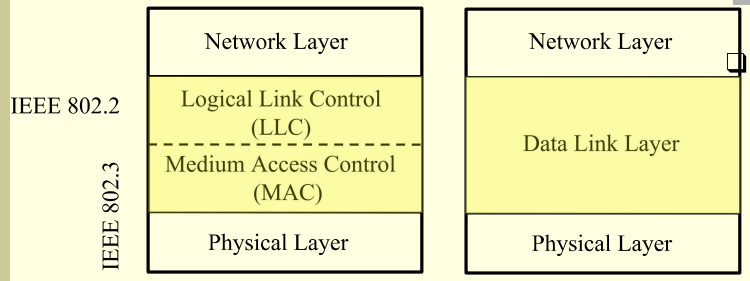
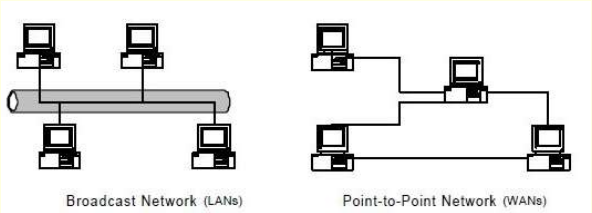
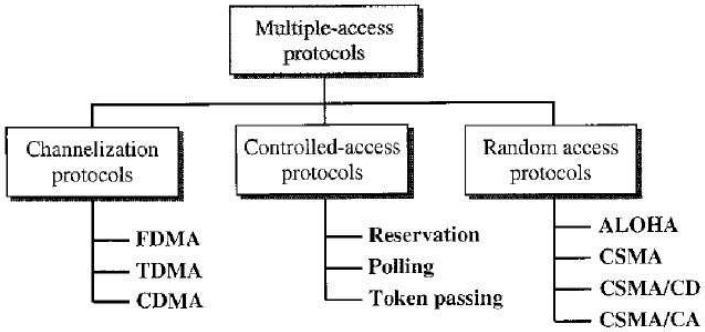
Components of Data Link Layer



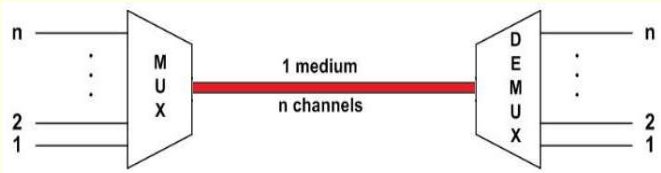
- LLC Functionality:
  - Framing
  - Flow Control
  - Error Control

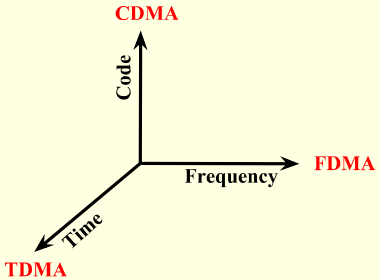


Various MAC Protocols

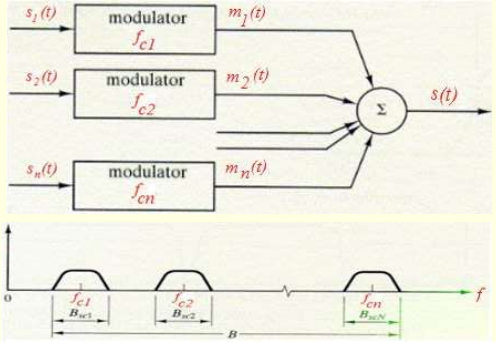


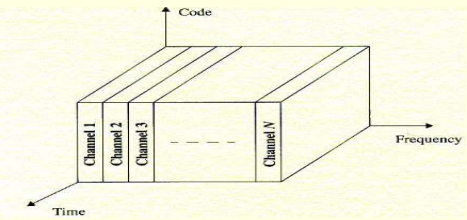
MAC based on Channelization Protocol

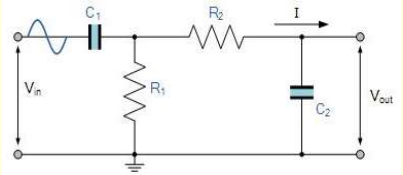




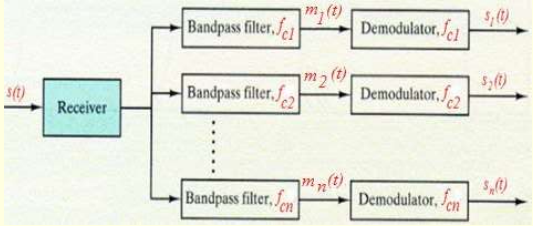
Frequency Division Multiple Access (FDMA)



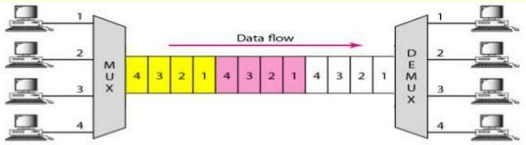


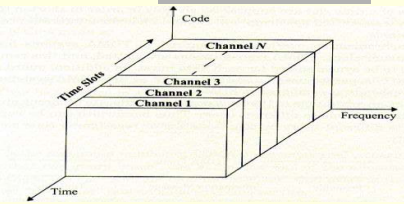


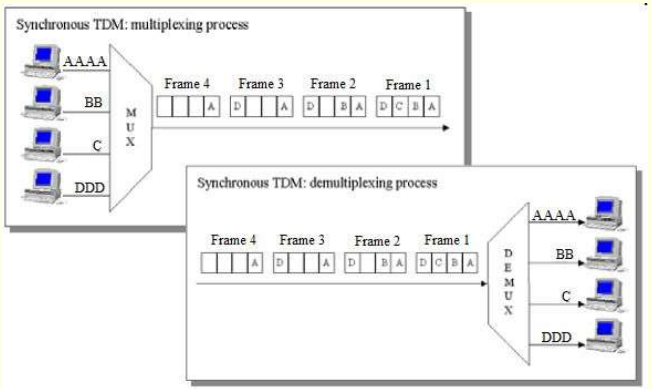
$f_L = \frac{1}{2\pi R_1 C_1} Hz$ $f_H = \frac{1}{2\pi R_2 C_2} Hz$



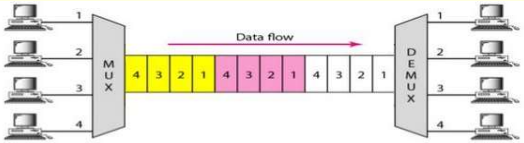
Time Division Multiple Access (TDMA)

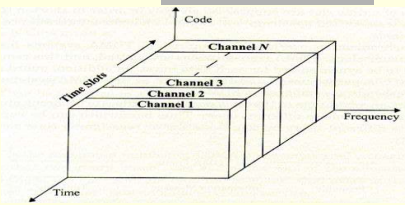


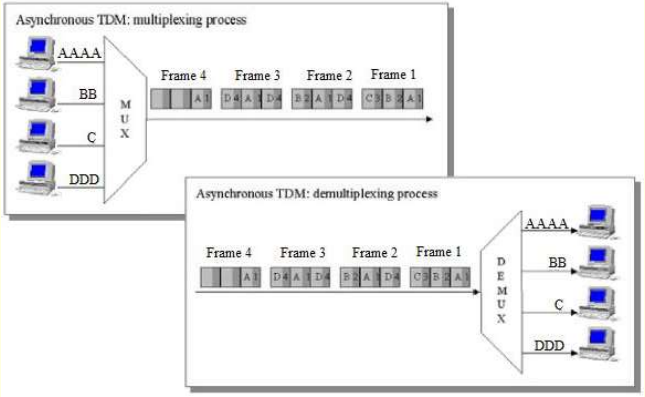




Time Division Multiple Access (TDMA)





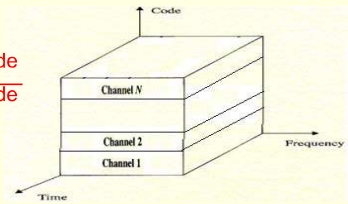


Code Division Multiple Access (CDMA)

A = [1 1 1 1]
B = [1 -1 1 -1]
C = [1 1 -1 -1]
D = [1 -1 -1 1]

Orthogonal Code :
$A.B = \sum A_i.B_i = 0$

Sending 0 □ Sending Code
Sending 1 □ Sending $\overline{\text{Code}}$



A □(0) C and B□ (1) C

A □ [1 1 1 1]
B □ [-1 1 -1 1]

C will receive
R □ [0 2 0 2]

At C:
A's Data = R.A = [0 2 0 2]. [1 1 1 1] = 4 □ 0
B's Data = R.B = [0 2 0 2]. [1 -1 1 -1] = -4 □ 1
D's Data = R.D = [0 2 0 2]. [1 -1 -1 1] = 0 □ No Data

Walse Code :

$$W_2 = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

$$W_{2n} = \begin{pmatrix} W_n & W_n \\ W_n & \overline{W_n} \end{pmatrix}$$

Problem of Static Allocation

- Arrival Rate λ frames/sec
- Bandwidth C bps
- Average Frame Length 1/μ bits i.e. Service Rate μC frames/sec
- As per M/M/1 Queuing Model, we have Mean Delay

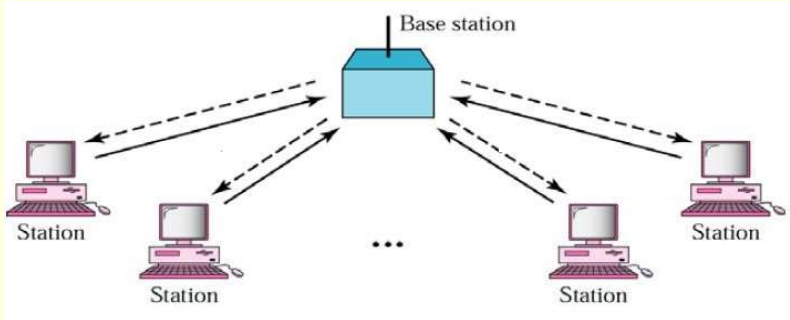
$$T = \frac{1}{\text{service rate} - \text{arrival rate}} = \frac{1}{\mu C - \lambda}$$

- If we divide the channel into N sub channels, each with capacity C/N bps, we have

$$T_N = \frac{1}{\mu(C/N) - (\lambda/N)} = \frac{N}{\mu C - \lambda} = NT$$

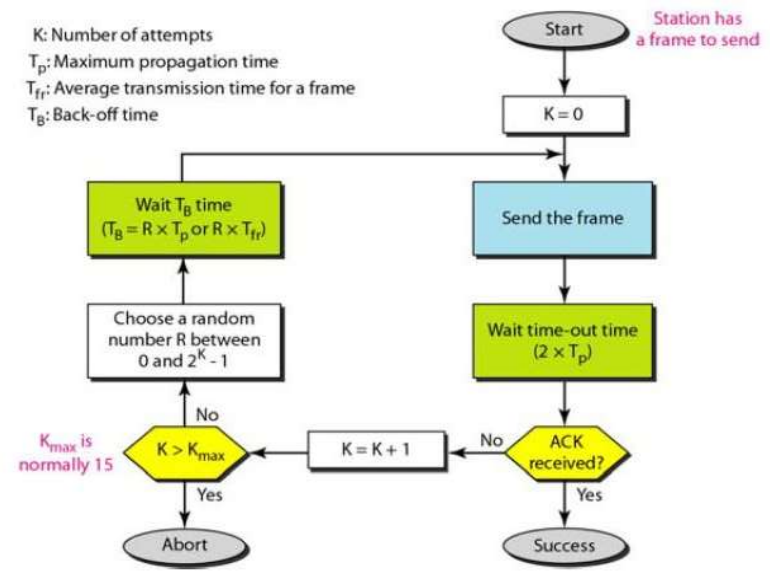
ALOHA Protocol

- Developed at University of Hawaii in 1970

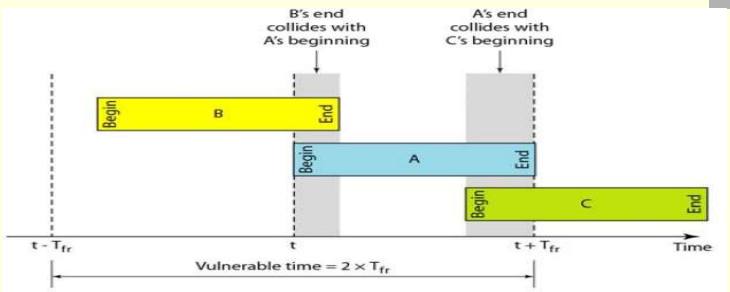


- Two Versions
  1. Pure ALOHA Time is continuous
  2. Slotted ALOHA Time is discrete

Pure ALOHA Protocol



Efficiency of Pure ALOHA Protocol

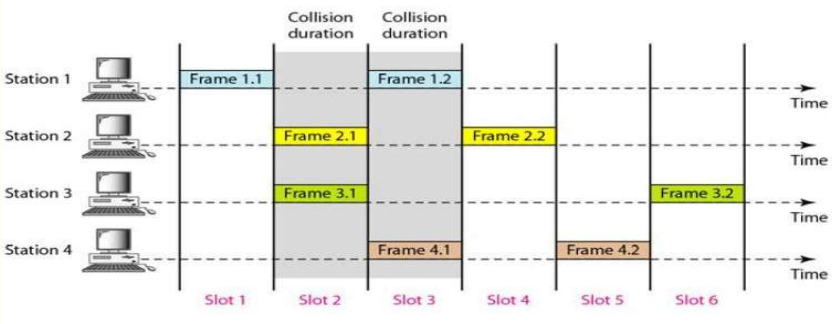


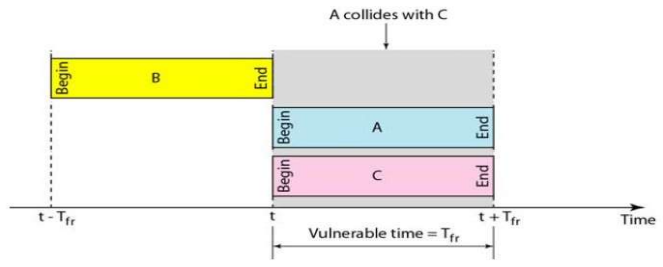
- G Average number of frames generated in the system during T_{fr}
- Throughput $S = G.P_0$
- Probability that there will be k frames during the vulnerable time is

$$P_k = \frac{(2G)^k e^{-2G}}{k!}$$

- Throughput $S = G.P_0 = G.e^{-2G}$
- S is maximized at G = 0.5 and $S_{max} = 1/2e = 0.184$
- Thus the maximum throughput of pure ALOHA = 18.4%

Slotted ALOHA Protocol



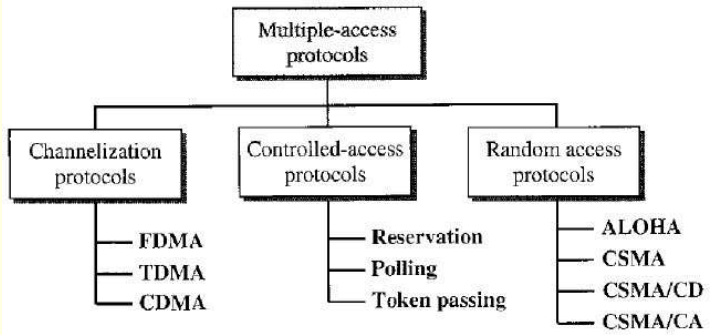


$$P_k = \frac{G^k e^{-G}}{k!}$$

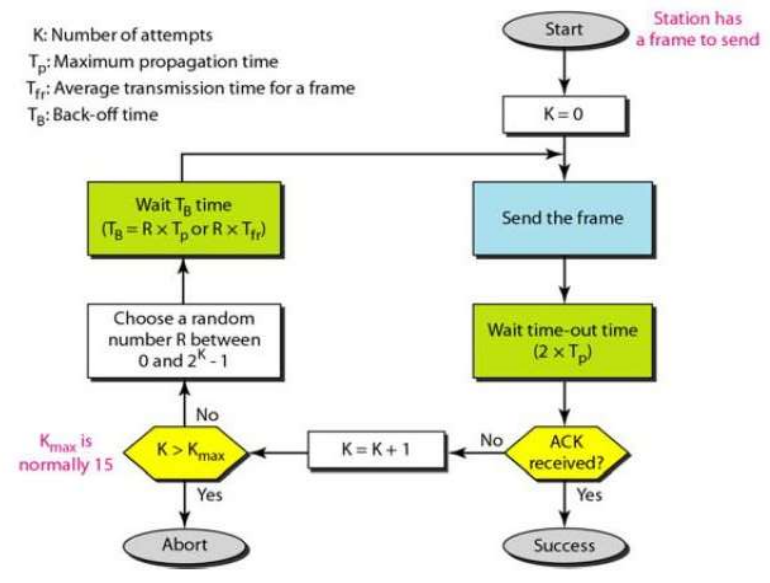
$$S = Ge^{-G}$$

$$S_{max} = 1/e = 0.368$$

Various MAC Protocols

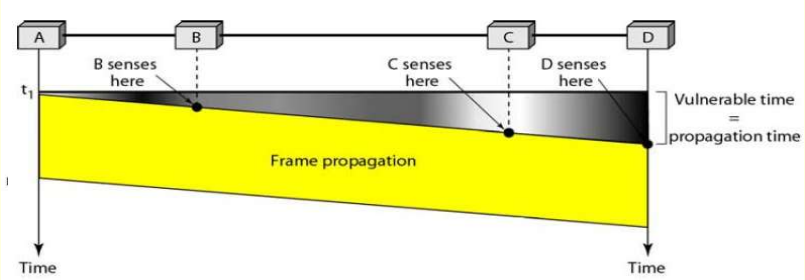


ALOHA Protocol



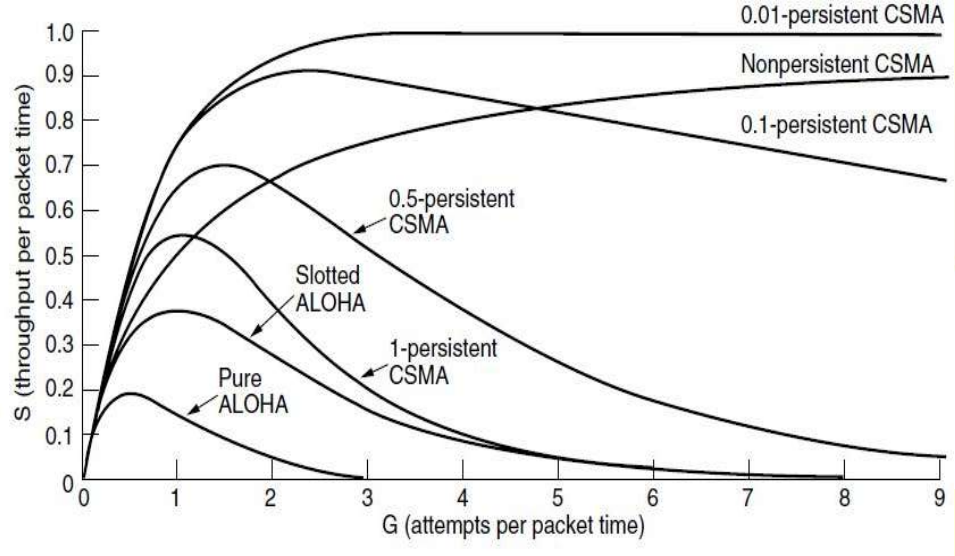
CSMA Protocol

- In ALOHA protocol, whenever a Station has a Frame it sends the Frame immediately. Thus the probability of collision is very high.
- In CSMA, before sending a Frame sense the medium for carrier (signal).
- Is there any chance of collision?
  - Both senses the medium at the same time.
  - Propagation Delay



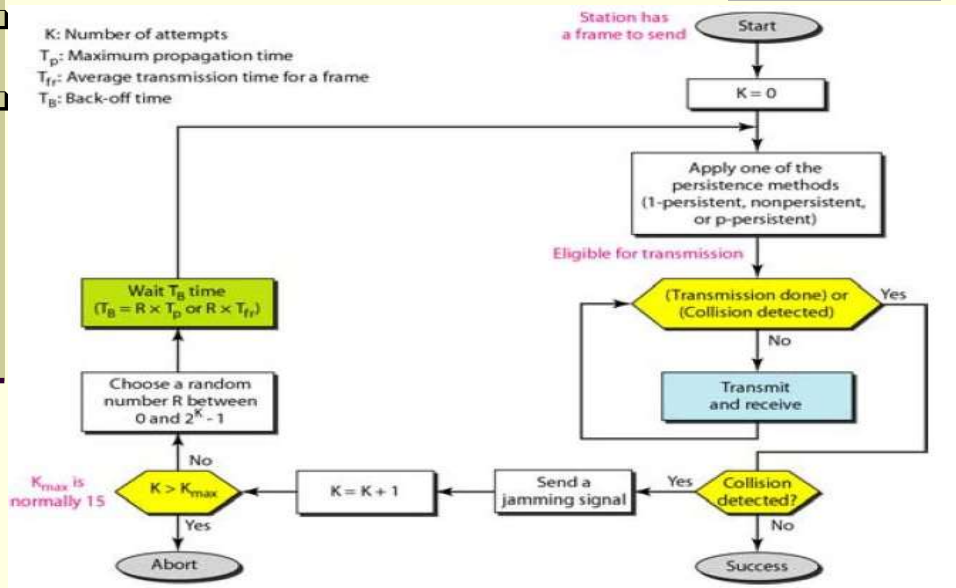
- What a station should do if the channel is idle or the channel is busy?
  - 1-persistent CSMA, Nonpersistent CSMA and p-persistent CSMA

CSMA Protocol



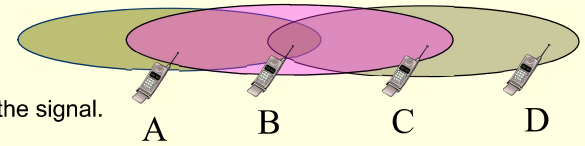
CSMA/CD Protocol

- K: Number of attempts
  T_p: Maximum propagation time
  T_{fr}: Average transmission time for a frame
  T_B: Back-off time



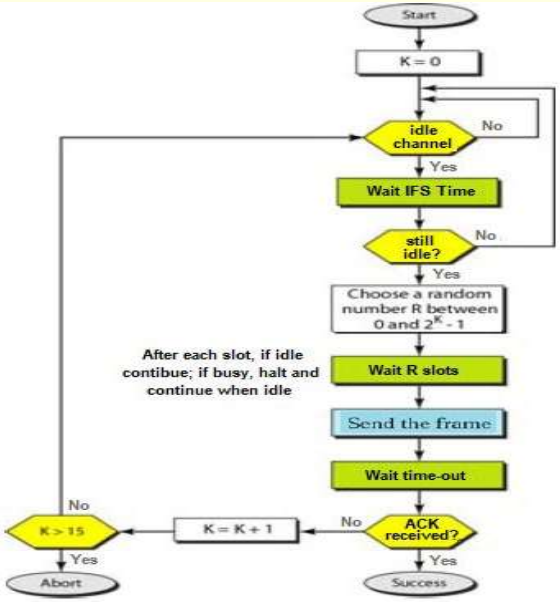
CSMA/CA Protocol

- Wireless network uses CSMA/CA protocol.
- Why not CSMA/CD for wireless network?
  - Hidden Terminal Problem:
    - A & C can't hear each other.
    - A sends to B, C can't retrieve the signal.
    - C wants to send to B, C senses a free medium (CS Fails).
    - Collision occurs at B.
    - A & C can't retrieve the collision(CD Fails).
    - C is "hidden" from A.



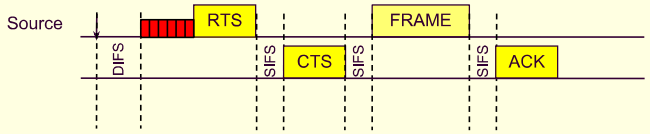
  - Exposed Terminal problem:
    - B sends to A, C can retrieve the signal.
    - C wants to send data to D
    - C senses a busy medium and will wait unnecessarily till B A ends.
    - B is "exposed" to C

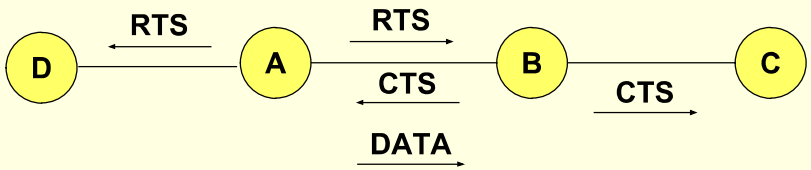
CSMA/CA Protocol



CSMA/CA Protocol

- Has the previous solution solved Hidden Terminal Problem or Exposed Terminal Problem?
  - RTS(Request To Send) & CTS(Clear To Send) frames, prior to actual data frame, are sent.





- Hidden Terminal C overhears CTS while Exposed Terminal D overhears RTS.